

HaloNet: Physics-Anchored Pseudo-Label Generation for Box-Supervised Infrared Small Target Detection

Input: IR Image + Bounding Box

1 Problem

2 B-SNR Task

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3 PAG

2 PAG Soft Label [Group D Output]

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2 PAG Soft Label [Group D Output]

4 Model-Agnostic

5 Results

3 Model-Agnostic Training

Plug any IRSTD backbone:

DNANet

(TIP 22)

Dense nested UNet++

ACM

(WACV 21)

Asymmetric non-local

ALCNet

(TGRS 21)

Local contrast FPN

↓ SoftIoULoss (sigmoid applied internally)

Training with PAG soft labels
→ pixel-level segmentation

(*) No architecture modification
(*) No extra parameters

The Annotation Bottleneck

- [x] Pixel-level mask: expensive & slow
- [x] Single-point label: hard to click accurately on dim 3x3 px targets
- [v] Bounding Box: auto-output by YOLO or radar systems

STEP 1 Estimate background stats from Context Box (1.5x expanded): μ_{ctx} , σ_{ctx} (background only)

STEP 2 Compute per-pixel posterior:

$$W(i) = \sigma(\tau \cdot (I(i) - \mu_{ctx}) / \sigma_{ctx})$$

→ High W = likely TARGET pixel
→ Low W = likely BACKGROUND pixel

Derived from Bayesian posterior under Gaussian background model

STEP 3 Find physical hotspot: $p^* = \text{argmax}(W) \leftarrow$ NOT box center!

STEP 4 FWHM anchor region: $\Omega = \{i : W(i) \geq W(p^*) \times 0.5\}$

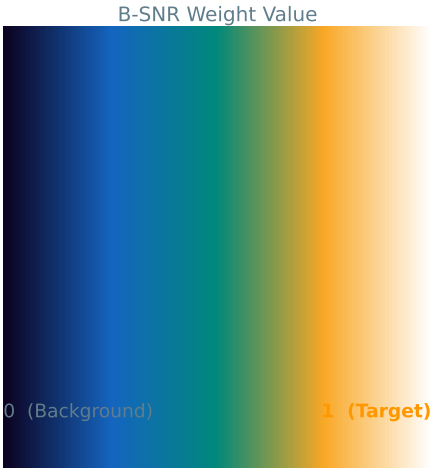
STEP 5 SNR-weighted spatial spread: $\sigma = 1.5 \times \sqrt{(\sum_{\Omega} W(i) \cdot d_i^2) / \sum_{\Omega} W(i)}$

STEP 6 Physics-Anchored Gaussian: $P(i) = W(i) \times \exp(-d_i^2 / 2\sigma^2)$

→ σ tracks target's physical radiance spread
→ Robust to box position errors

PAG vs B-SNR

Why PAG > B-SNR:
B-SNR captures SNR shape but may retain background ripples.
PAG multiplies with a Gaussian anchored to the physical energy peak → clean, compact, smooth soft label that better matches target PSF.



[!] Theoretical Limitation (Variance Corruption Bound): $\hat{\sigma}_{ctx}^2 \approx \sigma_o^2 + \alpha(1-\alpha) \cdot \Delta\mu^2 \rightarrow$ When target-to-box ratio $\alpha < 0.05$ (ultra-small targets in NUDT-SIRST), background variance is inflated by target brightness, causing SNR $\rightarrow 0$. This formally explains the performance gap on NUDT.

4 Experimental Results (mIoU %)

DNANet backbone · 3 datasets

Dataset	LESPS	PAL	C (B-SNR)	D (PAG)	GT (Full)
NUAA-SIRST	55.17	63.46	63.12	70.69	72.84
NUDT-SIRST	57.39	73.10	56.12	62.97	95.07
IRSTD-1K	50.90	60.72	60.91	60.75	70.94
Average	54.49	65.76	60.05	64.80	79.62

B-SNR (C) PAG (D) ★

Model-Agnostic Verification (NUDT, ~1000 ep):

Backbone	GT (A)	PAG (D)	Retention
ALCNet	65.76%	55.81%	84.9%

64.85% 53.37% 82.3%

>> PAG > B-SNR consistently across ALL backbones